Stage 1–3 Machine Access Checklist & Re-scoped Test Plan

> **Purpose:** the machines (and middleware PCs) we need for bench/integration testing across Stages 1–3 of `LIS_IMPLEMENTATION_PLAN.md`, cross-checked against the `manuals-and-lis-protocol` knowledgebase (KB), and **re-scoped against the units we actually have**.
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> **Status:** drafted 2026-06-26 · **availability received 2026-06-26.**
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> **Headline:** the available fleet **covers all three stages** — but **5 of the 7 available units are not the machines the plan originally named.** The plan's Stage-1 targets (RAC-050, "Mindray labXpert") and most Stage-2/3 targets are **not on hand**, so we substitute available HL7-equivalents. Net effect: a **stronger, multi-vendor Stage 1**, a **thinner Stage 2** (one confirmed-ASTM unit), and a **Stage 3 gated on middleware** (SnibeLis PC + license).

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0. TL;DR — re-scoped stages

Stage	Goal	Primary vehicle (available)	Ready?
1 — HL7/MLLP "first result"	one MLLP listener + HL7 parser, first normalized result	**EDAN H60S** (HL7 v2.4 / MLLP / port 7999)	✅ **unit in warehouse; matches existing edge (LIS-**
2 — ASTM / serial	ASTM E1381/E1394 stack	**ERBA EC90** (ASTM, RS232 or Ethernet)	✅ **unit in warehouse**
3 — proprietary tail	middleware-brokered result + QC	**SNIBE MAGLUMI X3** + **SnibeLis**	🟡 **analyzer on hand; blocked on SnibeLis PC + lic**

Bonus coverage now possible: HETO AU120 gives a *second-vendor* HL7/MLLP unit for Stage 1 (proves the parser isn't EDAN-specific); **RAYTO RT-7600** is a second proprietary-TCP tail for Stage-3 reverse-mapping practice.

The good news: we can now exercise Stage 1 across **three vendors** (EDAN confirmed, HETO likely, Seamaty/RAYTO provisional) — better vendor-tolerance proof than the original RAC-050-only plan. **The gap:** Stage 2 rests on a **single, upload-only ASTM unit** (ERBA EC90); the richest ASTM spec (DiaSys 920) is not available.

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1. Available machines — verified against the KB

Availability tiers from your 2026-06-26 status: **on hand** · **in warehouse** (retrieve) · **incoming** (~next week).

Machine	Availability	Type	Documented interface (KB)	Stage fit	Readiness / blocker
EDAN H60S	In warehouse	Hematology (CBC)	**HL7 v2.4 over MLLP/TCP**, **port 7999**, bidirectional; analyzer = **TCP client**, LIS listens. ✅ clean.	**1**	✅ **Best Stage-1 target.** Retrieve f
ERBA EC90	In warehouse	Electrolyte / ISE	**ASTM E1381 + E1394**; RS232 **or** Ethernet; **unidirectional** upload. ✅	**2**	✅ **Only confirmed-ASTM unit ava**
SNIBE MAGLUMI X3	On hand	Immunoassay (CLIA)	**ASTM E1394 via SnibeLis** middleware; bidirectional + **QC upload** ✅	**3**	🟡 **Needs SnibeLis PC + vendor l**
HETO AU120	Incoming (~next wk)	Clinical chemistry	**HL7 v2.3.1 (doc also says v2.5) over MLLP/TCP**, bidirectional. ⚠️ **family doc only** — examples use **AU400**, no AU120-specific doc.	**1**	🟡 **Confirm on arrival the AU120**
RAYTO RT-7600	On hand	Hematology (CBC)	❌ **No HL7/ASTM in docs.** Proprietary **TCP "Netport"** record stream (vendor LIS-sim, port configurable) **or** RS232 (115200/8/N/1). Bidirectional-capable.	**3** (prov.)	🟡 **Byte-capture a "Send" to iden**
Seamaty SD1	On hand	Dry/whole-blood biochem	⚠️ **No SD1 doc in KB** — only the **SG1** (a *different* instrument: handheld blood-gas/ISE). SG1 = HL7 "v2.1.3" over TCP/serial, upload-only.	**1** (prov.)	🔴 **Obtain an SD1-specific LIS sp**
EDAN H99S	On hand	Unknown (likely hematology by name)	❌ **Not in KB at all** — no EDAN/H99S, not a recognizable EDAN catalog model. (Documented EDAN siblings: H60S hematology, I15 blood-gas, M16 immunoassay.)	?	🔴 **Read the unit nameplate/SN to**

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2. Re-scoped Stage 1–3 plan

Stage 1 — *first result through the pipe* → EDAN H60S

HL7 v2.4 over MLLP, analyzer dials our listener on **TCP 7999**, ACK returned — a textbook fit for the **MLLP de-frame + HL7 ACK^R01** transport already merged (LIS-13/S1.1). The H60S replaces the unavailable RAC-050 as the "first result" vehicle; it's a hematology ORU^R01 instead of a coagulation one, but the pipe is identical.

- **Access to confirm:** retrieve H60S from warehouse · RJ45 cable + switch/crossover · our edge listening on 7999 · firmware APP V1.10+.
- **Stretch (same stage, second vendor): HETO AU120** on arrival → prove the parser ingests a different vendor's HL7/MLLP without code changes. (Confirm AU120 = the AU-series HL7 screen; pin the port.)
- Provisional thirds: **Seamaty SD1** and **RAYTO RT-7600** *iff* their wire format turns out HL7 (both need a capture/vendor doc first).
- 📄 EDAN H60S LIS-Communication-Protocol p.7 (MLLP, TCP client, ACK) · LIS Connection Training p.4 (port 7999)

Stage 2 — ASTM / serial → ERBA EC90

The only confirmed-ASTM unit on the list. Exercises the ASTM E1381 low-level framing (ENQ/ACK/checksum) + E1394 records — the core Stage-2 deliverable. **Caveat:** it's **upload-only** (no host-query) and the richest bidirectional ASTM spec (DiaSys 920) isn't available, so Stage-2 bench breadth is reduced to one analyzer.

- **Access to confirm:** retrieve EC90 from warehouse · decide **RS232 vs Ethernet** path (both supported) · for RS232: DB9 cable + USB-serial adapter (capture undocumented baud/pinout on bench) · an ASTM-speaking host that ACKs the frames.
- **Coverage gap to decide:** is one upload-only ASTM unit enough to call Stage 2 "done," or do we source a second ASTM analyzer (e.g. a DiaSys RESPONS) for a bidirectional / NAK-retransmit negative test? (See Action #5.)

- 📄 ERBA EC90 LIS Interface V1.01 p.4 (ASTM E1381/E1394)

Stage 3 — proprietary tail → **SNIBE MAGLUMI X3 + SnibeLis**

Analyzer ↔ **SnibeLis middleware PC** speaks ASTM E1394 (TCP or RS232); SnibeLis is the LIS client and also flags QC (Westgard). The analyzer is on hand — **the work and the risk are the middleware**, not the instrument.

- **Access to confirm (the real blockers):** a **SnibeLis/SnibeLinker Windows PC** · **SnibeLis install + vendor activation** (machine-code → Reg-Code from SNIBE) · confirm how SnibeLis **forwards to OpenELIS** (relay vs DB/export — undocumented in KB; ask SNIBE).
- **Second tail (bonus): RAYTO RT-7600**'s proprietary TCP "Netport" stream is another reverse-mapping exercise for the same Stage-3 muscle.
- 📄 SnibeLis LIS User Manual V1.1 App.A (ASTM E1394) p.73 · Guidance of LIS p.3 (TCP/IP mode + Upload QC)

3. Action items / gaps to close

1. **Retrieve from warehouse:** EDAN H60S (Stage 1) and ERBA EC90 (Stage 2) are the two primary vehicles and are both in the warehouse — pull them first.
2. **Stage 3 unblock (critical path):** source a **SnibeLis PC**, get the **vendor license** (machine-code→Reg-Code), and ask SNIBE how SnibeLis **forwards results to OpenELIS**. Without this, MAGLUMI X3 can't be tested past the SnibeLis boundary.
3. **EDAN H99S — disambiguate:** read the physical **nameplate / serial** (it's not a documented EDAN model). If it's really an H60S/M16/I15, we already have docs; otherwise request the LIS spec from EDAN.
4. **Seamaty SD1 — get the doc:** the KB only has the **SG1** (different instrument). Obtain an **SD1-specific LIS/communication spec** from Seamaty; meanwhile photograph the SD1's LIS setup screen + capture a frame.
5. **HETO AU120 — confirm on arrival:** verify the AU120 exposes the same **HETO AU-series HL7/MLLP** interface (docs only show AU400); pin the **TCP port + listener role** (the manual omits them); confirm the **2.3.1-vs-2.5** wire version.
6. **RAYTO RT-7600 — byte-capture:** run a "Send" into a raw TCP listener (Netport, e.g. port 9102) **and** a serial capture (115200/8/N/1) to settle whether the wire format is HL7, ASTM, or proprietary — then file it into Stage 1/2/3.
7. **Stage 2 breadth decision:** decide whether **ERBA EC90 alone** (upload-only ASTM) satisfies Stage 2, or whether to acquire a **bidirectional ASTM** analyzer for the NAK-retransmit / host-query negative tests.
8. **Per-unit conformance:** each unit still needs the standard "supported" gate — protocol doc → simulator fixture → parser tests → bench-conformance (direction + sample + raw) → E2E green.

4. Availability of the plan's originally-named machines

For traceability — what the plan named vs what's reachable:

Plan machine	Stage	Available?	Substitute
RAYTO RAC-050	1	❌ No	**EDAN H60S (HL7/MLLP)**
"Mindray labXpert"	1	❌ No (and it's middleware, not a machine)	**HETO AU120 / EDAN H60S**
DiaSys RESPONS 920	2	❌ No	***(none — ASTM breadth gap)***
ERBA EC90	2	✅ **Yes** (warehouse)	**— (is the vehicle)**
GOLDSITE GPP-100	2	❌ No	**—**
HETO Konig AP300	2	🟡 AU120 incoming (sibling, HL7 not ASTM)	**HETO AU120 (→ Stage 1)**
MEDICA EasyStat	2	❌ No	**—**
HORRON EA2000	2	❌ No	**—**
SNIBE MAGLUMI (any)	3	✅ **Yes** — **X3** on hand	**— (is the vehicle)**
Mindray BC + DMS	3	❌ No	**RAYTO RT-7600 (other proprietary tail)**

Appendix A — original cross-check of the plan-named machines (reference)

Kept for context; this is what drove the substitutions above. **Two findings still matter:** (a) **"Mindray labXpert" is middleware, not an analyzer** — its real physical unit was never pinned; (b) the plan's **Stage-2 "ASTM/serial fleet" was mostly mislabeled** — GOLDSITE, HETO and HORRON are actually HL7/Ethernet, and MEDICA EasyStat is a proprietary text dump.

Plan machine	Stage	Real documented interface (KB)	Note
RAYTO RAC-050	1	HL7 v2.3.1 / MLLP / TCP (client, port 2000), bidir	✅ **as planned; ACK = original-only (not enhanced**
Mindray labXpert	1	**Middleware** (HL7 v2.3.1 or file-drop); unit unnamed	⚠️ **not a machine**
DiaSys RESPONS 920	2	ASTM E1381/E1394, RS232 or TCP:7777, bidir	✅ **ASTM; spec doc misfiled under RESPONS-940**
ERBA EC90	2	ASTM E1381/E1394, RS232 or Ethernet, upload-only	✅ **ASTM (now a primary vehicle)**
GOLDSITE GPP-100	2	**HL7 v2.3.1 / TCP / MLLP / port 8000**, bidir	❌ **not ASTM**
HETO Konig AP300	2	**HL7 / MLLP / Ethernet**, bidir	❌ **not ASTM (sibling of incoming AU120)**
MEDICA EasyStat	2	**Proprietary text dump, RS232 2400/8N1**, uni	❌ **not ASTM; needs cable Cat# 5637 + custom par**
HORRON EA2000	2	**HL7/TCP** native, or legacy serial+middleware	❌ **not ASTM; image-native PDF re-verify gate**
SNIBE MAGLUMI / SnibeLis	3	ASTM E1394 via SnibeLis PC, bidir + QC	✅ **as planned (X3 is the vehicle)**
Mindray BC / DMS	3	**HL7/Ethernet direct** (DMS optional; KB DMS = vet only)	❌ **DMS not required for modern BC**

Source citations (host/LIS docs)

Available units: EDAN H60S LIS Protocol · EDAN H60S LIS Training (port 7999) · ERBA EC90 LIS Interface V1.01 · SNIBE SnibeLis LIS User Manual V1.1 · HETO Konig LIS Protocol V2.1 · RAYTO RT-7600 LIS setup · RAYTO RT-7600 User Manual V1.4e (115200/8N1) · Seamaty SG1 manual (HL7 v2.1.3) · EDAN M16 HIS/LIS Interface (HL7 v2.4/MLLP/port 8000) *(EDAN H99S has no KB doc)*

Plan-named units (appendix): RAC-050, DiaSys 920, GOLDSITE GPP-100, MEDICA EasyStat, HORRON EA2000, Mindray BC — see prior revision / KB folders under each vendor. </content>